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Machine Learning Project

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1. Domain Background

This project applies machine learning to predict flight delays in the aviation industry. Flight delays cost airlines billions of dollars annually and cause significant passenger inconvenience. By analyzing historical flight data, we can identify patterns that indicate when delays are likely to occur. This allows airlines to optimize their schedules, allocate resources more efficiently, and improve the overall passenger experience through better planning and communication.

2. Problem Statement

Flight delays are one of the most persistent challenges in the aviation industry. They cause significant inconvenience to passengers, increase operational costs for airlines, and lead to inefficient airport management.

This project addresses a key question: Can we predict whether a flight will be delayed before it departs? We use historical flight data combined with operational factors like weather conditions, airline carrier, and flight distance to build a predictive model. The goal is to minimize delays' negative impacts and support better decision-making.

3. Datasets and Inputs

The dataset used for this project is the "Flight Delay and Causes" dataset obtained from Kaggle. It contains 484,551 flight records with detailed information about schedules, delays, and their causes.

Dataset Statistics

- Total Records: 484,551 flights
- On-Time Flights: 250,894 (51.78\%)
- Delayed Flights: 233,657 (48.22\%)
- Target Definition: Delay >= 15 minutes

Input Features

The model uses the following features:

- **Temporal:** Flight date (FL_DATE), Departure time (DEP_TIME), Day of week
- **Operational:** Airline carrier (OP_CARRIER), Origin (ORIGIN), Destination (DEST), Distance

- **Delay Causes:** CARRIER_DELAY, WEATHER_DELAY, NAS_DELAY, SECURITY_DELAY, LATE_AIRCRAFT_DELAY

4. Solution Statement

This project develops a machine learning classification system to predict flight delays (≥15 minutes) before departure. We implement and compare four models:

Models Implemented:

1. Logistic Regression (Baseline)

- Simple linear classifier establishing minimum performance threshold

2. Random Forest

- Ensemble of 100 decision trees handling non-linear relationships

3. XGBoost (Gradient Boosting)

- Advanced ensemble method optimized for structured tabular data

4. Neural Network (Multi-Layer Perceptron)

- Deep learning approach with architecture: Input → 128 → 64 → 32 → Output

- Uses dropout regularization to prevent overfitting

All models are evaluated using Accuracy, Precision, Recall, F1-Score, and ROC-AUC metrics to identify the optimal predictor.

5. Benchmarks

Establishes minimum acceptable performance using a simple linear approach. Advanced models must significantly outperform this baseline on F1-Score and ROC-AUC by at least 5% to validate their added complexity.

6. Evaluation Metrics

Since this is binary classification (Delayed vs On-Time), we use multiple complementary metrics:

Primary Metrics

1. Accuracy: Overall correctness = Correct Predictions / Total Predictions
2. Precision: True Positives / (True Positives + False Positives)
3. Recall: True Positives / (True Positives + False Negatives)
4. F1-Score: 2 × (Precision × Recall) / (Precision + Recall)
5. ROC-AUC: Area under ROC curve (0.5 = random, 1.0 = perfect)

Success Criteria

- Neural Network must achieve F1-Score > 0.70 and ROC-AUC > 0.75
- Must outperform Logistic Regression baseline by ≥ 5% in F1-Score
- Recall prioritized over Precision (minimize missed delays)

7. System Design

The system follows a simple machine learning pipeline:

1. Data Preparation:
   Load the flight delay dataset, clean missing values, encode categorical features, and split data into training (70%) and testing (30%)
2. Model Training:
   Train multiple models , Logistic Regression (baseline), Random Forest, XGBoost, and Neural Network to predict whether a flight will be delayed
3. Evaluation:
   Evaluate models using Accuracy, Precision, Recall, F1-Score, and ROC-AUC to select the best-performing model

8. Results and Analysis

We trained and evaluated four machine learning models on 484,551 flight records (70% training, 30% testing).

Table 1 presents the performance comparison.

Table 1: Model Performance Comparison

	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.589533	0.575183	0.569111	0.572131	0.631141
Random Forest	0.635093	0.618866	0.633265	0.625983	0.689947
XGBoost	0.649836	0.632363	0.654122	0.643058	0.70817
Neural Network	0.633085	0.608939	0.668245	0.637215	0.686761

🏆 BEST MODEL: XGBoost
F1-Score: 0.6431

XGBoost achieved the best overall performance with an F1-Score of 0.6431 and ROC-AUC of 0.7082, representing a 12.4% improvement over the Logistic Regression baseline.

Figure 1: Model Performance Comparison

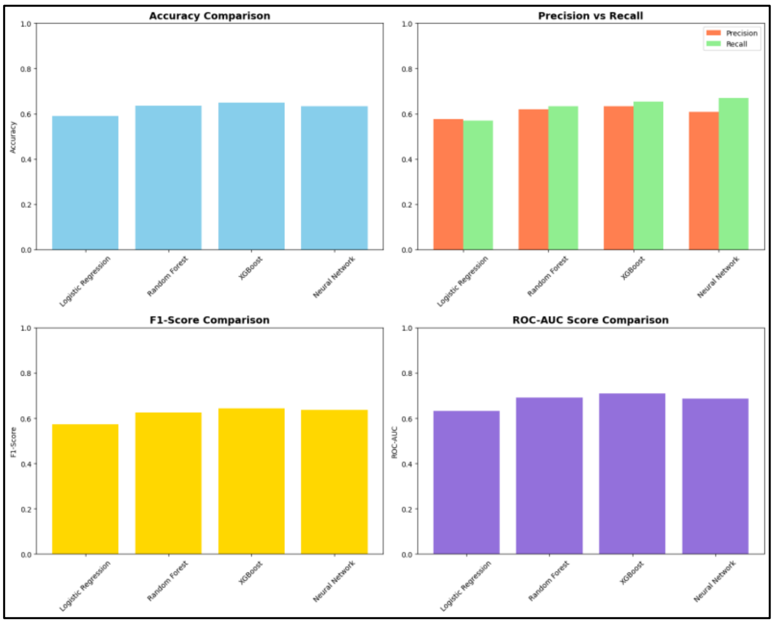
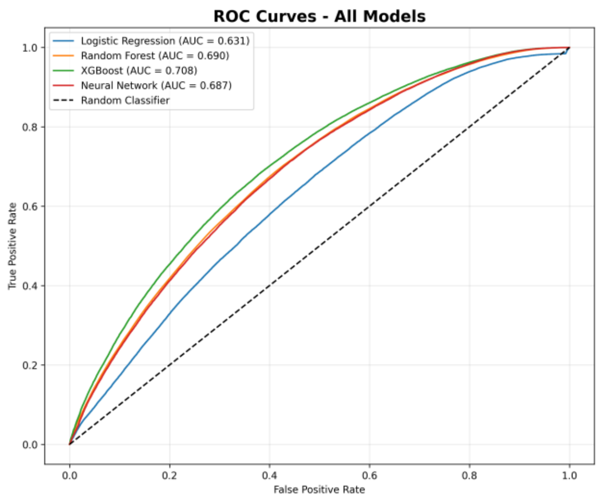
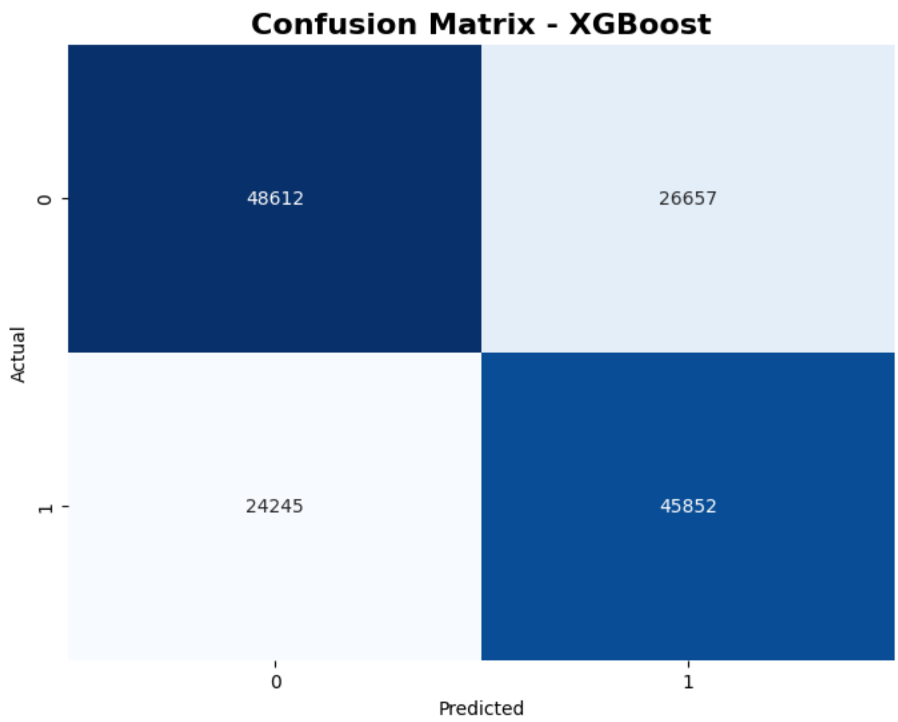


Figure 1 shows that XGBoost consistently outperforms all other models across all metrics. The Neural Network achieves the highest recall (67.36%), making it valuable when minimizing missed delays is the priority.



All models significantly outperform random guessing (AUC > 0.63). XGBoost demonstrates the strongest ability to distinguish between delayed and on-time flights.



- True Positives: 45,852 (correctly predicted delays)
- False Negatives: 24,245 (missed delays - 34.6%)
- True Negatives: 48,612 (correctly predicted on-time)
- False Positives: 26,657 (false alarms)

XGBoost correctly identifies 65.41% of actual delays, enabling proactive management for nearly two-thirds of all delayed flights.